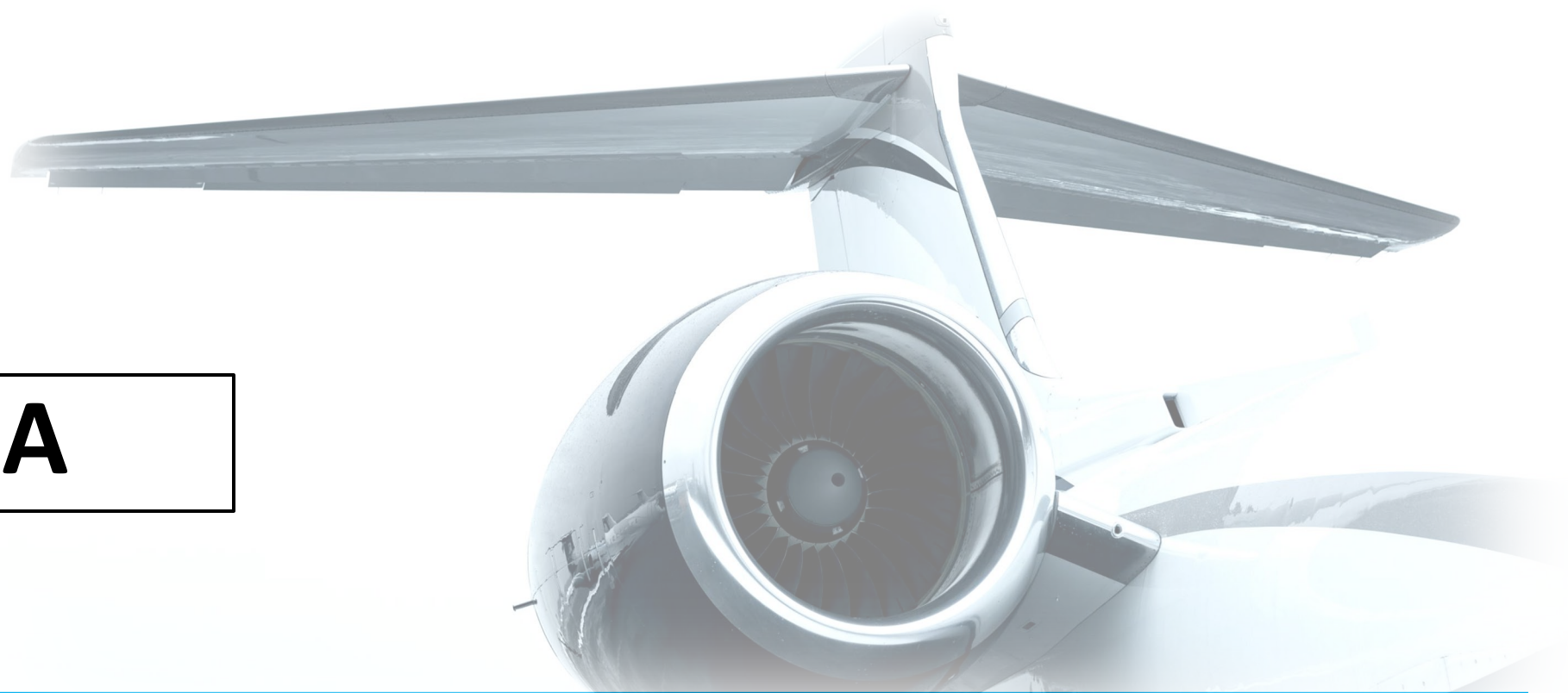


Safety and Reliability Training

Module 2: RELIABILITY ENGINEERING

Submodule 2.B :FMEA

Pin FMEA





Objectives of this training

- ✓ To grasp the necessities and difficulties of FMEA process
- ✓ To understand the Pin FMEA methodology and its advantages.





Outline

- ☐ Pin FMEA - Introduction
- ☐ Pin FMEA – Complex Parts/COTS
- ☐ Pin FMEA - Purpose
- ☐ Pin FMEA - Methodology
- ☐ The future



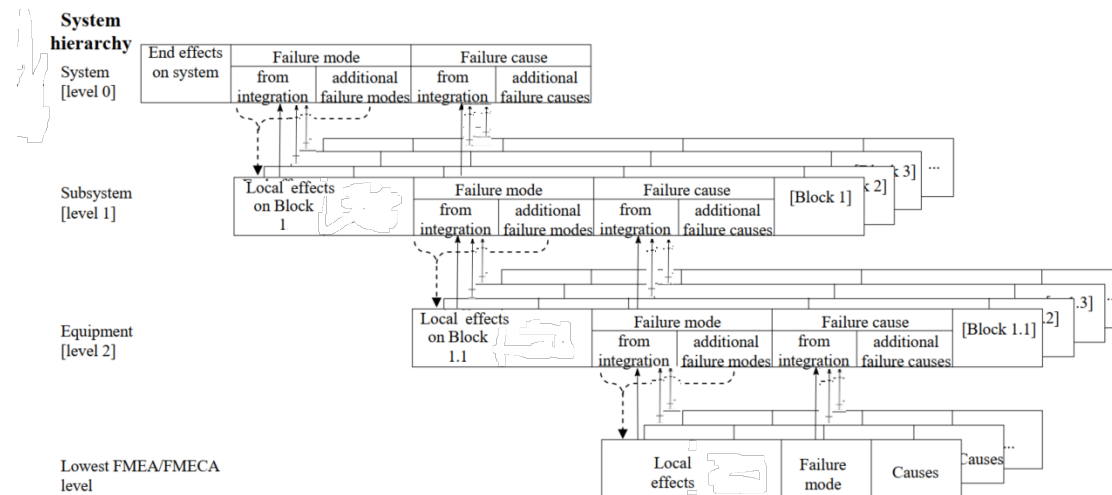
Pin FMEA – Introduction (I)

- What a Pin FMEA is?? Does it prick?



Pin FMEA – Introduction (II)

- The Failure Mode and Effect Analysis (FMEA) is a powerful RAM technique to evaluate systems and equipment inherent capability to fail and propagate failures.
- FMEA methodology is based on the identification of the potential failure modes of each item according to experience and engineering judgement, and their effects on the upper system levels.



Pin FMEA – Introduction (III)

- Failure Mode determination is of vital importance when performing a PPFMEA since all the analysis effects are affected.
- When dealing with simple Parts/COTS (resistors, capacitor, mechanical elements...etc.), their classification can provide generic failure modes. However, in the case of complex ones this task can be quite difficult because of the different functions considered and the component design (causes) variability.



Pin FMEA – Complex Parts/COTS (I)

- Microcircuits (also named as Integrated Circuits), even considered as a part category in Reliability standards or libraries, is a heterogeneous group which has not commonalities other than technology and packaging. Furthermore, part failure modes are dependent on their particular functionality.
- Microcircuit manufacturers seldomly provide the list of part failure modes and even more rarely their percentual distribution.
- Because of all reasons, the analysis can result in erroneous conclusions.



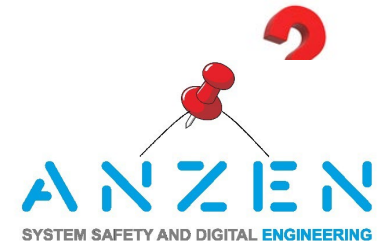
Pin FMEA – Purpose (I)

- A pin FMEA is a systematic technique, both qualitative and quantitative, for failure mode identification of complex Parts/COTS, which consider their specific functionalities.
- It is based on the estimation of their failures modes as a consequence of certain failures at pin-related level.
- It is just an approximation since it cannot consider all the possible causes of failure modes. However, results are more reliable than other methods based on component classification or libraries.



Pin FMEA – Purpose (II)

- By using Pin FMEAs:
 - Complex Parts/COTS are modelled in a more precise way which in turn benefits FMEA quality and time-consuming.
 - Failure mode libraries can be created this way, providing ANZEN with a valuable know-how which can be taken advantage of in different projects.



Pin FMEA - Methodology (I)

- Firstly, a clear picture of part functionality and pinout is to be obtained, for example from its datasheet.

TEXAS INSTRUMENTS
ISO7730, ISO7731
SLLS650J – SEPTEMBER 2016 – REVISED OCTOBER 2024

ISO773x High-Speed, Robust-EMC Reinforced and Basic Triple-Channel Digital Isolators

1 Features

- 100Mbps data rate
- Robust isolation barrier:
 - >30-year projected lifetime at 1500V_{RMS} working voltage
 - Up to 5000V_{RMS} isolation rating
 - Up to 12.8kV surge capability
 - ±100kV/μs typical CMTI
- Wide supply range: 2.25V to 5.5V
- 2.25V to 5.5V Level translation
- Default output high (ISO773x) and low (ISO773xF) options
- Wide temperature range: -55°C to +125°C
- Low power consumption, typical 1.5mA per channel at 1Mbps
- Low propagation delay: 11ns Typical (5V Supplies)
- Robust electromagnetic compatibility (EMC)
 - System-level ESD, EFT, and surge immunity
 - ±8 kV IEC 61000-4-2 contact discharge protection across isolation barrier
 - Low emissions
- Wide-SOIC (DW-16) and QSOP (DBQ-16) package options
- Automotive version available: [ISO773x-Q1](#)
- Safety-Related Certifications:**
 - DIN EN IEC 60747-17 (VDE 0884-17)
 - UL 1577 component recognition program
 - IEC 61010-1, IEC 62368-1, IEC60601-1, and GB 4943.1 certifications

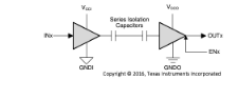
The ISO7730 device has all three channels in the same direction and the ISO7731 device has two forward and one reverse-direction channel. If the input power or signal is lost, the default output is high for devices without suffix F and low for devices with suffix F. See the *Device Functional Modes* section for further details.

Used in conjunction with isolated power supplies, this family of devices helps prevent noise currents on data buses, such as RS-485, RS-232, and CAN, or other circuits from entering the local ground and interfering with or damaging sensitive circuitry. Through innovative chip design and layout techniques, electromagnetic compatibility of the ISO773x device has been significantly enhanced to ease system-level ESD, EFT, surge, and emissions compliance. The ISO773x family of devices is available in 16-pin wide-SOIC and QSOP packages.

Package Information

PART NUMBER	PACKAGE ⁽¹⁾	PACKAGE SIZE ⁽²⁾
ISO7730	SOIC (DW)	10.30mm × 7.50mm
ISO7731	QSOP (DBQ)	4.90mm × 3.90mm
ISO7731B	SOIC (DW)	10.30mm × 7.50mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.



Simplified Schematic

V_{CC}=Input supply, V_{CCO}=Output supply
GND=Input ground, GND=Output ground

2 Applications

- Industrial automation
- Motor control
- Power supplies
- Solar inverters
- Medical equipment

3 Description

The ISO773x devices are high-performance, triple-channel digital isolators with 5000V_{RMS} (DW package) and 3000V_{RMS} (DBQ package) isolation ratings per UL 1577.

This family includes devices with reinforced insulation ratings according to VDE, CSA, TUV and CQC. The ISO7731B device is designed for applications that require basic insulation ratings only.

The ISO773x family of devices provides high electromagnetic immunity and low emissions at

TEXAS INSTRUMENTS
ISO7730, ISO7731
SLLS650J – SEPTEMBER 2016 – REVISED OCTOBER 2024

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4 Pin Configuration and Functions

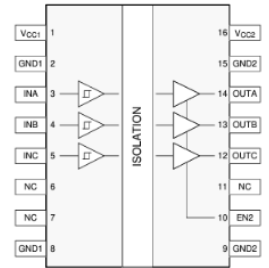


Figure 4-1. ISO7730 DW and DBQ Packages 16-Pin SOIC-WB and QSOP Top View



Pin FMEA - Methodology (II)

- For some microcircuits, a valuable Pin Failure Mode Analysis (Pin FMA) is provided by manufacturer.

Functional Safety Information

ISO7731/ISO7731-Q1 Functional Safety FIT Rate, FMD and Pin FMA



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4 Pin Failure Mode Analysis (Pin FMA)	6
4.1 16-QSOP (SSOP) and 16-DW (wide-body SOIC) Package	6

Trademarks

All trademarks are the property of their respective owners.



Failure Mode Distribution (FMD)

3 Failure Mode Distribution (FMD)

The failure mode distribution estimation for ISO7731/ISO7731-Q1 in Table 3-1 comes from the combination of common failure modes listed in standards such as IEC 61508 and ISO 26262, the ratio of sub-circuit function size and complexity and from best engineering judgment.

The failure modes listed in this section reflect random failure events and do not include failures due to misuse or overstress.

Table 3-1. Die Failure Modes and Distribution

Die Failure Modes	Failure Mode Distribution (%)
OUT state undetermined	35%
OUT not in timing or voltage specification	30%
OUT stuck to default state	25%
OUT stuck high	5%
OUT stuck low	5%

The FMD in Table 3-1 excludes short circuit faults across the isolation barrier. Faults for short circuit across the isolation barrier can be excluded according to ISO 61800-5-2:2016 if the following requirements are fulfilled:

1. The signal isolation component is OVC III according to IEC 61800-5-1. If a SELV/PELV power supply is used, pollution degree 2/OVC II applies. All requirements of IEC 61800-5-1:2007, 4.3.6 apply.
2. Measures are taken to ensure that an internal failure of the signal isolation component cannot result in excessive temperature of its insulating material.

Creepage and clearance requirements should be applied according to the specific equipment isolation standards of an application. Care should be taken to maintain the creepage and clearance distance of a board design to ensure that the mounting pads of the isolator on the printed-circuit board do not reduce this distance.



Pin FMEA methodology (II)

- Then, with the aid of a [Pin FMEA spreadsheet template](#), each of the integrated circuit pins is submitted to four generic failures:
 - Open
 - Shortcircuited to power supply
 - Shortcircuited to ground
 - Shortcircuited to next pin
- Estimated effects at microcircuit level are included and described.
- Any relevant data should be added for the sake of analysis traceability.



Pin FMEA methodology (III)

- Finally, the spreadsheet summarize all the equal effects to reduce their quantity to a few ones.
- The relative frequency of the resulting effects are automatically calculated to generate the final part failure mode distribution.
- After revision, the results can be incorporated to a ANZEN library and to the corresponding analyses.



Pin FMEA methodology (IV)

P/N COMP.	FABRICANTE	LINK	PIN	PIN TYPE	SIGNAL	DESCRIPCION PIN/SIGNAL	MODOS FALLO PIN (=CAUSA MODOS FALLO COMP.)	EFFECTO FALLO PIN (=MODOS FALLO COMP.)	COMENTARIOS	FINAL EFECTOS (M.F. COMP.)	FINAL FMD. COMP. (%)
ISO7731FQDBQ	Texas	https://www.winsupport.ti.com/es/product-detailed/texas-instruments/iso7731q-dbaq1.html	PIN1		1 VCC1	Supply Pin	Open	Unpredictable or incorrect output		Unpredictable or incorrect output	7,94%
							Shorted to supply	N/A		Power supply shorted	13,60%
							Shorted to GND	Power supply shorted		No operation	9,83%
							Shorted to next pin	N/A		Logic low output	17,37%
			PIN2		2 GND1	Ground Pin	Open	No operation		No output	34,36%
							Shorted to supply	Power supply shorted		Logic high output	11,71%
							Shorted to GND	N/A		Logic high output if inputs are	11,71%
							Shorted to next pin	Logic low output		Undue internal synchronization	6,05%
			PIN3		3 INA	Logic Input. Depending on the version and POL bit setting, rising or falling edges on IN will propagate to OUT after a programmable delay. The LTC6994-1 will delay only the rising or falling edge. The LTC6994-2 will delay both edges.	Open	No output		No external enablement	11,71%
							Shorted to supply	No output			
							Shorted to GND	No output			
							Shorted to next pin	No output			
			PIN4		4 INB	Logic Input. Depending on the version and POL bit setting, rising or falling edges on IN will propagate to OUT after a programmable delay. The LTC6994-1 will delay only the rising or falling edge. The LTC6994-2 will delay both edges.	Open	No output			
							Shorted to supply	No output			
							Shorted to GND	No output			
							Shorted to next pin	No output			
			PIN5		5 OUTC	Output	Open	No output			
							Shorted to supply	Logic high output			
							Shorted to GND	Logic low output			
							Shorted to next pin	Logic high output if inputs are			
			PIN6		6 SYNC	External synchronization input pin. Applying an external clock to this pin forces synchronization of the internal oscillator to the external clock frequency for low frequency designs (<520 kHz). If this pin is not used, it has an internal pull-down resistor so it can be left open or connected to ground.	Open	Undue internal synchronization			
							Shorted to supply	N/A			
							Shorted to GND	No effect			
							Shorted to next pin	No effect			
							Open	No external enablement			



The future

- Currently, we have a trained procedure in Chat GPT for accelerating Pin FMEAs process. Results are yet to be validated.

Pepito... Pin FMEA of LT4356HDE-1, pins TMR, FB, OUT, GATE, SNS, VCC, SHDN, FLT, EN, GND, AOUT, IN+, EPAD

Here's the Pin FMEA for the LT4356HDE-1 with the specified pins. The table includes the failure modes, effects, summarized effects, and their relative frequencies.

Pin FMEA: LT4356HDE-1

Pin	Failure Mode	Effect	Summarized Effect	Relative Frequency (%)
TMR	Open	Timer disabled; overcurrent protection compromised.	Overcurrent protection disabled	6.9%
	Short to VCC	Timer causes early fault triggering.	Overcurrent protection disabled	6.9%
	Short to GND	Fault timeout disabled; improper fault protection.	Overcurrent protection disabled	6.9%
FB	Open	Output voltage regulation lost; possible overvoltage.	Incorrect voltage regulation	6.9%
	Short to VCC	Output voltage increases uncontrollably.	Incorrect voltage regulation	6.9%
	Short to GND	Output voltage drops; load may shut down.	Incorrect voltage regulation	6.9%
OUT	Open	No voltage to the load; system shutdown.	Device does not function	10.34%
	Short to VCC	Output exceeds expected levels; damage to load possible.	Incorrect voltage regulation	6.9%
	Short to GND	No voltage supplied; potential damage due to high current.	Device does not function	10.34%
GATE	Open	MOSFET control disabled; poor current regulation.	Improper MOSFET control	6.9%
	Short to VCC	MOSFET remains on; constant current flow.	Improper MOSFET control	6.9%
	Short to GND	MOSFET remains off; no current to load.	Improper MOSFET control	6.9%
SNS	Open	Sense input disconnected; poor overcurrent protection.	Overcurrent protection disabled	6.9%
	Short to VCC	Incorrect sense input; false triggering.	Overcurrent protection disabled	6.9%

	VCC		function	
AOUT	Open	Auxiliary signal not generated; incorrect feedback.	Loss of monitoring or feedback.	6.9%
	Short to VCC	Signal stuck high; false signal to external circuit.	Loss of monitoring or feedback.	6.9%
	Short to GND	Signal stuck low; false signal to external circuit.	Loss of monitoring or feedback.	6.9%
IN+	Open	Input disconnected; improper or no regulation.	Incorrect voltage regulation.	6.9%
	Short to VCC	Output unregulated; unsafe.	Incorrect voltage regulation.	6.9%
	Short to GND	Input low; output shuts down or becomes unstable.	Incorrect voltage regulation.	6.9%
EPAD	Open	Poor thermal dissipation; overheating risk.	Thermal issues (overheating)	6.9%
	Short to VCC	Device malfunction; potential damage.	Device does not function.	10.34%
	Short to GND	Normal operation.	No negative effect.	-

Summary of Effects:

- Device does not function: 24.14%
- Incorrect voltage regulation: 20.69%
- Overcurrent protection disabled: 13.79%
- Fault signal failure: 13.79%
- Improper MOSFET control: 10.34%
- Loss of monitoring or feedback: 10.34%
- Thermal issues (overheating): 6.9%

This table now includes the summarized effects and their relative frequencies. If you need more details or further adjustments, just let me know!

🗨️ 📄 📌 🔄 📧



Thank you!!!

